

DCVCP: Experimental Evaluation of Leukaemia Disease Prediction using Deep Convoluted Vector Classification Principle

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Abstract- A wide variety of cancers, including skin, breast, lung, and blood malignancies such as lymphoma and leukemia, are defined by the uncontrolled growth of cells within the body. Acquired lymphoblastic leukemia is among the most prominent cancers. Blood cancer diagnoses are notoriously time-consuming because hematologists often make mistakes. With the use of cutting-edge tools like Deep Learning and Machine Learning, this study considers a fresh approach to leukemia classification. The precise diagnosis of blood cancer is a crucial medical process, but it is also one of the most costly and challenging for clinical staff to do. Blood cancer is one of many severe forms of the disease. The devastating and deadly blood cancers, lymphoma and leukemia, impact people of all ages and genders, and they are a leading cause of other illnesses and an elevated survival rate. Damage to and an increase in immature monocytes, neutrophils, eosinophils, and lymphocytes are hallmarks of both leukemia and lymphoma. Predicting and treating blood cancer early is crucial for improving survival rates in the health industry. These days, a number of manual methods exist for analyzing and predicting blood cancer based on microscopic medical reports of white blood cell images; this method is both reliable and deadly. It takes a lot of time and effort to manually forecast and analyze eosinophils, lymphocytes, monocytes, and neutrophils. The Deep Convoluted Vector Classification Principle (DCVCP) is a new deep learning technique for leukemia illness detection that was described in this study. To test how well it works, it was cross-validated with the current method, the Convolutional Neural Network (CNN). findings from this study show that DCVCP can effectively leverage data from network sensors to identify blood cancer, which might lead to faster findings and lower costs in clinical settings.

Index Terms—Blood Cancer, Leukemia Disease, Deep Convoluted Learning, Vector Classification, DCVCP, Convolutional Neural Network, CNN.

I. INTRODUCTION

The phrase "blood cancer" is used to describe a group of malignancies that impact the lymph nodes, bone marrow, and blood itself. Multiple myeloma, lymphoma, and leukemia subtypes are included in this group of malignancies [1][2]. The symptoms of blood cancer can differ substantially from person to person, making diagnosis a challenge. But typical symptoms include being run down, hot all the time, sweating at night, losing weight for no apparent reason, lymph nodes that swell, and becoming sick again and again. Hence, keep an eye out for these symptoms and get medical help if you see any of them getting worse. The key to early detection of blood cancer is an accurate diagnosis. In order to provide effective treatment and results, a blood cancer diagnosis is a crucial first step. Doctors are able to respond and give patients the best treatment possible when they notice conditions early on [3][4]. Here are some key points from our research:

- (i) **Early Detection:** Diagnosing blood cancer at an early stage improves treatment outcomes, lessens the impact of symptoms, and decreases the likelihood of death.
- (ii) **Precise Screening:** By checking a patient at regular intervals, doctors can learn more about their health and detect cancer in its early stages.
- (iii) **Effective Treatment:** When doctors know the type of blood cancer a patient has, they can provide them the best possible care [5][6].

Blood cancer cells metastasize to vital organs and tissues all throughout the body by way of the circulatory system, lungs, brain, and arteries. Typically, the job of delivering oxygen from the heart to every organ in the body is done by red blood cells [7]. Half of the blood volume is

composed of these. However, white blood cells are crucial to human immunity and act as the body's initial defense against many different types of illness. Therefore, in order to comprehend the symptoms of the disease, it is necessary to precisely identify these white blood cells [8]. Anemia of the acute lymphoblastic kind arises from alterations in lymphocytes, a type of white blood cell. Leukemia can be either acute or persistent. Although chronic leukemia takes longer to manifest than acute leukemia, the usual recovery period for acute myeloid leukemia without therapy is around three months. Nearly a quarter of all childhood cancers are chronic lymphoblastic leukemia, making it the most prevalent form of acute leukemia. Hematologists, researchers, and physicians have long struggled with the problem of early leukemia and lymphoma detection [9][10]. Leukemia cells quickly multiply in the blood and then move to other parts of the body, including the liver, spleen, lymph nodes, brain, and neurological system. This process can cause a broad variety of symptoms. Bruising, nose and gum bleeding, fever, enlarged lymph nodes, painful joints, and infections are among these symptoms. Acute lymphoblastic leukemia mostly affects the blood and bone marrow. The prevalence of this condition in children led to its characterization as "acute childhood leukemia" rather than "chronic" or "myeloid" leukemia. Early detection increases the likelihood of a cure, but unchecked, it can be fatal within months.

II. RELATED STUDY

Cancers of the bone marrow and blood, such as leukemia, impact the formation of blood cells [11]. Impure cells, particularly white blood cells, have deoxyribonucleic acid (DNA) damaged by this disease. A radiologist must be involved in the lengthy and accurate process of diagnosing acute leukemia cells. Consequently, a number of research projects have been undertaken with the goal of creating algorithms for the automated diagnosis of acute leukemia. Nevertheless, the detection speed and accuracy of these investigations are rather poor. Nowadays, medical research relies heavily on machine learning and AI methods, especially for the detection and classification of leukemic cells. By assisting with earlier illness detection, these technologies lessen doctors' workload and mistake potential. The goal of this research is to create a deep learning model that, with the help of a tailored architecture, can detect acute leukemia in images of monocytes and lymphocytes. Pictures of AML and ALL (Acute Lymphoblastic Leukemia) are shown in this new collection. To aid researchers in their pursuit of medical research that makes use of machine learning techniques, a new dataset has been assembled with the aid of a number of specialists. Generative Adversarial Networks (GANs) provide for dataset scalability. Suggestions for convolutional neural network (CNN) models for acute leukemia image classification have six convolution layers, four dense layers, as well as a Softmax activation function

based on the Tversky loss function. The suggested approach was able to diagnose ALL and AML, two forms of acute leukemia, with a 99% accuracy rate. The suggested network outperforms the state-of-the-art in terms of accuracy and speed, and the findings suggest that this model might be useful in assisting medical professionals in real-world settings [11].

An important part of identifying cancer and many other disorders is looking at peripheral blood smears [12]. Leukemia is a malignancy that starts in the bone marrow and causes an overabundance of white blood cells in the bloodstream. Leukemia is diagnosed when white blood cells exhibit unusual abnormalities in appearance. Using an image processing methodology, this research proposes an automated method for leukemia identification. From peripheral blood smears stained with Leishman, 1159 pictures were obtained in this investigation, spanning a range of brightness levels and color hues. The white blood cell samples were sorted into normal and pathological groups using a support vector machine (SVM) classifier. It was also used to detect aberrant white blood cells (WBCs) that were leukemic. Five subgroups of normal white blood cells were identified using NN classifier. A classification precision of 98.8 percent was achieved by combining NN with SVM [12].

A key to effectively treating blood cancer, which has been on the rise over the past decade, is a prompt diagnosis [13]. A diagnosis requires a team of medical specialists and a battery of tests, all of which add time and money to the procedure. For this reason, a reliable automated diagnostic system is required for creating forecasts. The use of machine learning techniques to the analysis of leukemia micro-array gene data for the purpose of blood cancer detection has recently assumed more significance. A lot of research has gone into this, but it could all be more efficient and accurate. Predicting the incidence of blood cancer using supervised machine learning is proposed in this research. This study incorporates the 22,283-gene leukemia microarray gene set. We implement ADASYN resampling as well as Chi-squared (Chi2) features selection algorithms to address imbalanced and high-dimensional dataset concerns. Chi2 selects 22,283 features for use in training learning models, whereas ADASYN generates fictitious data to level the dataset for each classification. Our logistic vector trees classifier, LVTrees, combines logistic regression, support vector classifier, and extra tree classifier for classification. In addition to conducting thorough tests on the datasets, we have compared the performance of the suggested methodology to that of state-of-the-art approaches to establish its importance. Using ADASYN and Chi2 methods, LVTrees achieve a remarkable 100% accuracy rate, surpassing all other models. Also, to prove that the strategy works, a T-test for statistical significance is run. The suggested model is superior, according to results obtained by k-fold cross-validation [13].

Leukemia is a deadly disease that puts a lot of people at risk of dying. Recognizing it early can greatly enhance its remission rate [14]. Instead of using traditional methods, which have a number of drawbacks, this study suggests two automated classification models that use transfer learning and are based on pictures of blood microscopically to identify leukemia. In the first model, pictures of blood microscopy are pre-processed. Next, a deep convolutional neural network called AlexNet, which has been trained using several popular classifiers, is used to extract features. Following picture pre-processing in the second model, AlexNet is trained to extract features and classify them. The findings demonstrated that the second model performed better than the first owing to its 100% classification accuracy, using a dataset of 28, 20 images for the testing. [14].

Acute leukemia is one type of cancer that can arise from problems in the bone marrow [15]. Teens and young adults seem to get it more often. Rapid initiation of treatment is necessary since this leukemia type causes abnormal cell proliferation in a very short time. People afflicted with this malignancy mostly succumb to Acute Myeloid Leukemia (AML) and Acute Lymphoid Leukemia (ALL). Importantly, an automated approach can help clinicians choose the right treatment by classifying these two leukemia kinds on pictures of blood slides. We provide convolutional neural network (CNN) architecture in this research that can differentiate among ALL, AML, and normal blood samples. A precision of 97.23% and an accuracy of 97.18% were attained in the trials that used 16 datasets including 2,415 photos. We compared the outcomes of the proposed model to those of state-of-the-art approaches, some of which relied on convolutional neural networks (CNNs) [15].

III. METHODOLOGY

Numerous individuals are afflicted with blood cancer annually, which is a deadly disease. Diagnosing blood cancer can be a time-consuming and difficult process, unfortunately. Medical personnel must have a thorough understanding of blood cancer and its effects on the body in order to identify and treat the illness successfully. The patient's doctor may also request imaging tests like X-rays, CT scans, or MRI to gain a clearer picture of the patient's internal anatomy. Cancer can be diagnosed if the biopsy finds a large number of aberrant cells. The patient may remain at risk for future development of the illness even if the biopsy does not reveal any cancer cells. Leukemia, lymphoma, and myeloma are the three most common forms of blood cancer. It is crucial for doctors to accurately classify cancer types to avoid needless complications from treating the improper kind. Diagnosing cancer in the blood is a complex matter that calls for an integrated strategy. To start, doctors and nurses should be well-versed on the many different types of cancer, including those that affect the blood and others

that do not. Numerous tests can be performed to ascertain the kind and stage of blood cancer after it has been detected or is believed to exist. Different types and stages of cancer dictate different treatment choices.

Typical methods of treatment involve chemo, radiotherapy, and/or surgical procedures. Problems specific to each treatment option do exist. This entails diligently keeping note of any symptoms that may emerge, as well as having blood tests done on a regular basis. Following these procedures can help in monitoring and managing treatment-related side effects and problems.

When compared to the current technique, Convolutional Neural Network (CNN), the new method, Deep Convolved Vector Classification Principle (DCVCP), gives practitioners a suitable way to determine if a patient has acute leukemia. In order to halt the progression of childhood cancer, the study aims to create methods for quick and precise cell identification. This technique detected each and every cell on the blood smears sample slide by employing DCVCP. This research primarily aims to enhance the tool available to doctors in hospital hematology laboratories for DCVCP cell detection. The following figure Fig.1 shows the system architecture.

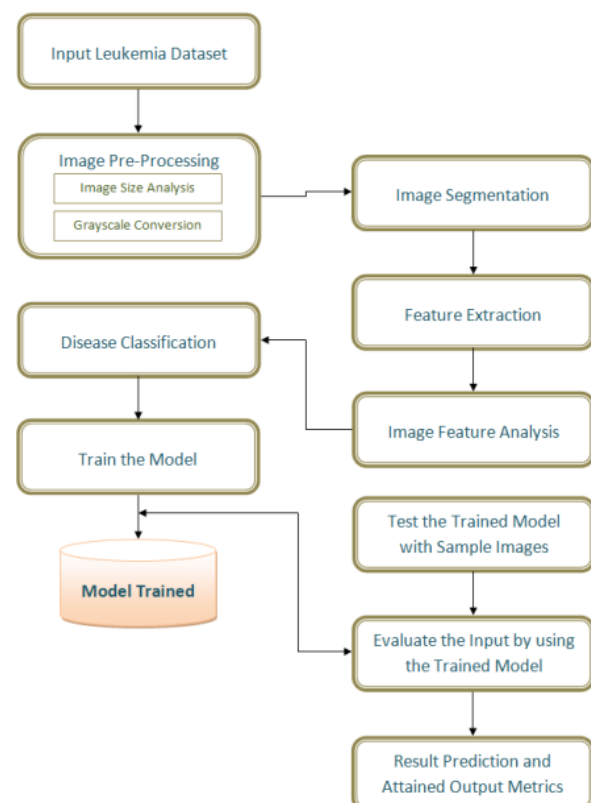


Fig.1 System Architecture

(i) Collecting Information: On the leukemia dataset, we test the suggested model's ability to predict the likelihood of leukemia malignancy. The dataset was sourced from

the Kaggle repository and is openly available to the public. Kaggle is a data science platform where users may upload and search datasets, work with other data scientists and publishers in the field of machine learning and deep learning, and build and test models in a variety of settings.

(ii) Preprocessing of Datasets: Before developing a DCVCP model for illness prediction, data must be preprocessed. The purpose of data preprocessing is to eliminate the noisy input that is causing the model's performance to degrade. There are unwanted regions and blank spots in every single picture in a leukemia dataset. Accordingly, cropping the photographs is critical for removing extraneous details and making better use of the available area.

(iii) Extracting Features: Following the cropping of all photos in the leukemia dataset, the subsequent step is to extract features. The numerical representation of the dataset employed is feature extraction. The dataset is prepared for DCVCP model application by transforming all raw data into relevant information.

(iv) Data Scaling: When aspects of a dataset are accessible on many scales, it is crucial to assign them to a scale equivalent. The features are rescaled to give them the characteristics of a normal distribution as part of the standardization procedure.

(v) Feature Selection: It is quite important to pick features during the preprocessing phase of data analysis. For intelligent systems to attain high accuracy, features must give enough characteristics to classify data into groups or classes for further inference and decision-making.

IV.RESULTS AND DISCUSSIONS

Gathering information from various hospital sources is the first step in the proposed approach. Data augmentation approaches were employed to overcome the problems of mining data samples for white blood cell (WBC) classifications after all samples of WBC had been collected. The four main phases in the entire process of blood cancer prediction are illustrated by the suggested technique. Phase one of the proposed frameworks involves gathering data from the hospital as well as pre-processed blood samples. These samples are then divided into training and testing ratios and stored in a confidential location so they may be easily accessed at any stage. In the training phase, data samples for classification and CNN and DCVCP algorithms are loaded from the private cloud.

The training logics include square root of the signal propagation, momentum estimation, and gradient descent classification. The model is trained using all the algorithms. Then, after applying learning criterion procedures, each algorithm's server stores the trained model individually. If the learning criteria satisfy the expectations of the proposed framework, then the model is ready for further review. Apply image processing

techniques like histogram equalization and retrain the model to ensure it meets the learning criterion if it fails the first time. Figure 2 shows the input image for the suggested strategy.

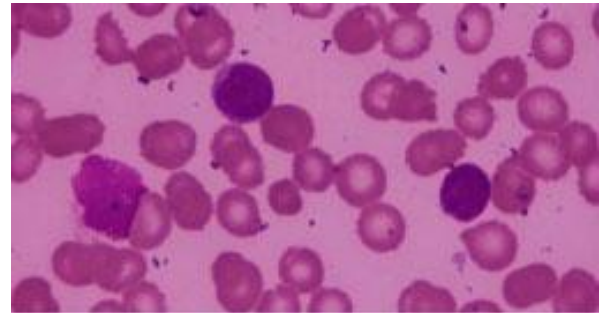


Fig.2 Input Image

The results of the suggested scheme's Pre-Processing, Segmentation, Feature Extraction, and Classification stages are shown in Figures 3, 4, 5, and 6.

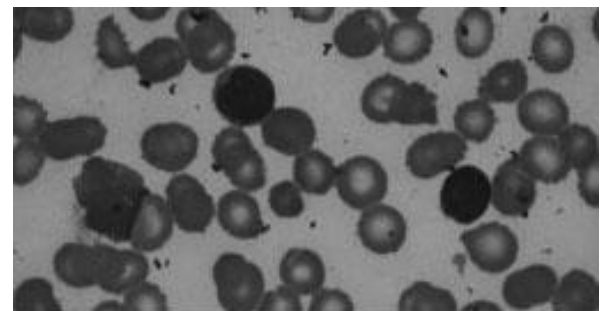


Fig.3 Pre-Processing

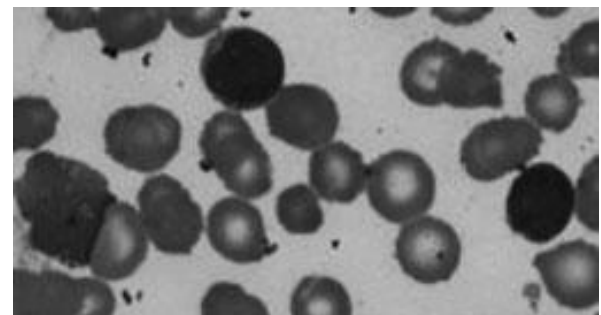


Fig.4 Segmentation

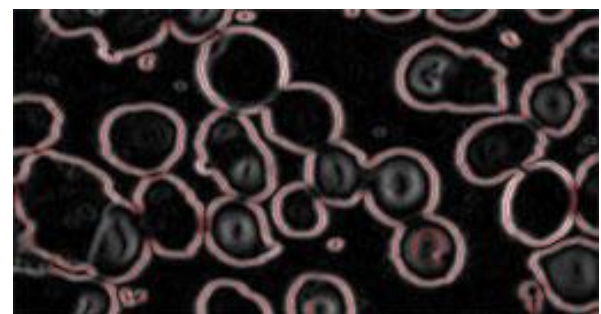


Fig.5 Feature Extraction

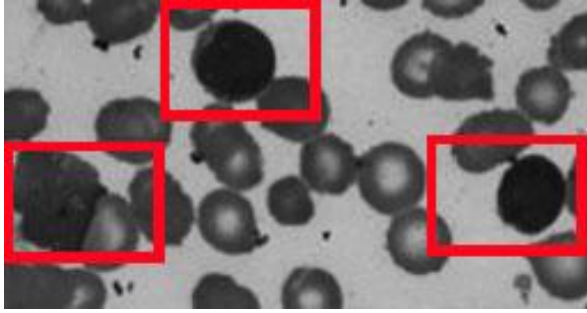


Fig.6 Classification

To evaluate the proposed model, we can see in Fig. 7 the accuracy of prediction of the DCVCP scheme when it is cross-validated using the CNN conventional learning method. Table-1 provides a descriptive representation of the same.

Table-1: Prediction Accuracy Comparison between DCVCP and CNN

Iterations	CNN (%)	DCVCP (%)
100	92.65	96.98
200	92.74	97.09
300	92.36	96.54
400	91.47	96.73
500	90.56	96.49
600	92.77	96.36
700	91.09	96.23
800	92.77	96.10
900	91.59	96.86
1000	91.43	96.48

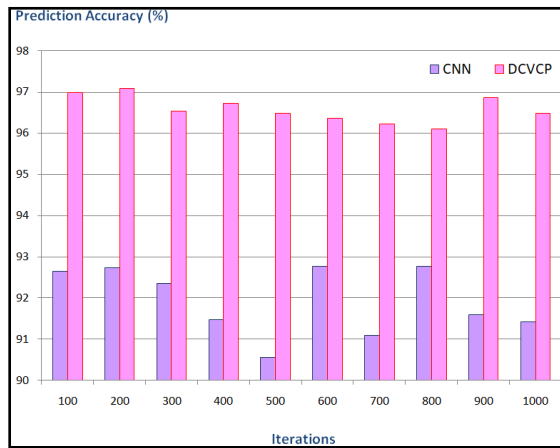


Fig.7 Prediction Accuracy

The time efficiency of the DCVCP scheme, that is cross-validated using the CNN conventional learning approach, is shown in the accompanying figure (Fig. 8) so that the performance of the recommended model may be evaluated. Table-2 provides a descriptive representation of the same.

Table-2: Comparison of Time Efficiency between DCVCP and CNN

Iterations	CNN (%)	DCVCP (%)
100	86.54	97.25
200	87.25	97.52
300	86.42	97.61
400	85.71	96.62
500	86.29	97.12
600	85.83	96.63
700	85.63	96.60
800	85.42	96.44
900	85.22	96.29
1000	85.01	96.14

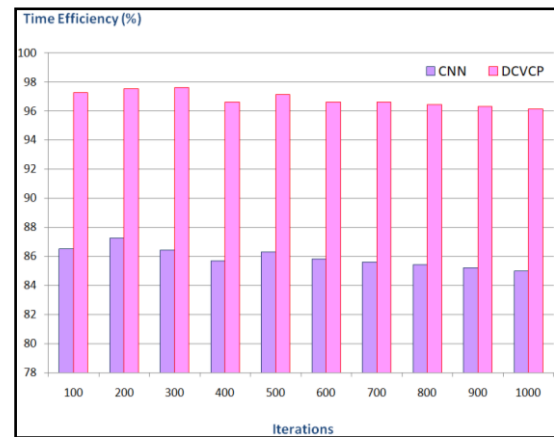


Fig.8 Time Efficiency

V.CONCLUSION

Despite a number of challenges for future study, Deep Convolutional Vector Classification Principle (DCVCP) has shown to be a diverse, accurate, and dependable diagnostic tool. Cancer of the blood is among the worst diseases a person may have. Neglected, it can cause the body to spread, destroy healthy cells, and trigger the unchecked production of white blood cells, which can be fatal. Leukemia is a terrible form of blood cancer that affects a lot of individuals. It occurs in the bone marrow in the same way as abnormal, uncontrolled multiplication of white blood cells causes' cancer. Leukemia, multiple myeloma, Hodgkin's lymphoma, and non-lymphoma are all examples of hematological malignancies, which consist of tumors with a link to blood. Leukemia affects all age groups, from adolescents to young adults and beyond, with the risk increasing with each passing year. Acute myeloid leukaemia (AML), chronic myelogenous leukaemia (CML), and acute lymphoblastic leukaemia (ALL) are all components of this category (CLL). Furthermore, we have selected the DCVCP model that has been used thus far and conducted comparative research using a systematic review to ascertain the technique that is most suitable for implementation. The future diagnosis and treatment of leukemia and other blood malignancies have been planned and developed by scientists and researchers.

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